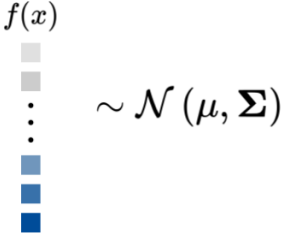
Introduction to Machine Learning

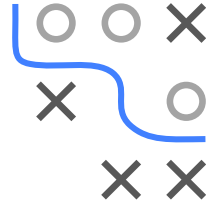
Gaussian Processes
Basics



Learning goals

- GPs model distributions over functions
- The marginalization property makes this distribution easily tractable
- GPs are fully specified by mean and covariance function
- GPs are indexed families

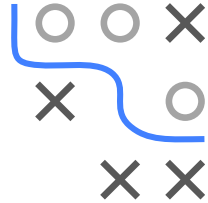
WEIGHT-SPACE VIEW



- Until now we considered a hypothesis space $\mathcal{H}$ of parameterized functions $f(\mathbf{x} \mid \boldsymbol{\theta})$ (in particular, the space of linear functions).
- Using Bayesian inference, we derived distributions for $\boldsymbol{\theta}$ after having observed data $\mathcal{D}$.
- Prior believes about the parameter are expressed via a prior distribution $q(\boldsymbol{\theta})$, which is updated according to Bayes' rule

$$\underbrace{p(\boldsymbol{\theta}|\mathbf{X},\mathbf{y})}_{\text{posterior}} = \frac{\overbrace{p(\mathbf{y}|\mathbf{X},\boldsymbol{\theta})}^{\text{likelihood}}\,\overbrace{q(\boldsymbol{\theta})}^{\text{prior}}}{\underbrace{p(\mathbf{y}|\mathbf{X})}_{\text{marginal}}}.$$

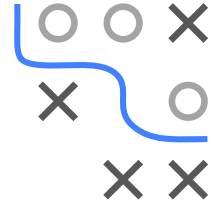
FUNCTION-SPACE VIEW



Let us change our point of view:

- Instead of “searching” for a parameter $\boldsymbol{\theta}$ in the parameter space, we directly search in a space of “allowed” functions $\mathcal{H}$.
- We still use Bayesian inference, but instead specifying a prior distribution over a parameter, we specify a prior distribution **over functions** and update it according to the data points we have observed.

WEIGHT-SPACE VS. FUNCTION-SPACE VIEW

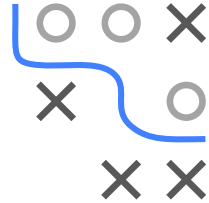


Weight-Space View	**Function-Space View**
Parameterize functions Example: $f(\mathbf{x} \mid \boldsymbol{\theta}) = \boldsymbol{\theta}^\top \mathbf{x}$	
Define distributions on $\boldsymbol{\theta}$	Define distributions on f
Inference in parameter space Θ	Inference in function space $\mathcal{H}$

Next, we will see how we can define distributions over functions mathematically.

Distributions on Functions

DISCRETE FUNCTIONS



For simplicity, let us consider functions with finite domains first.

Let $\mathcal{X} = \left\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right\}$ be a finite set of elements and $\mathcal{H}$ the set of all functions from $\mathcal{X} \to \mathbb{R}$.

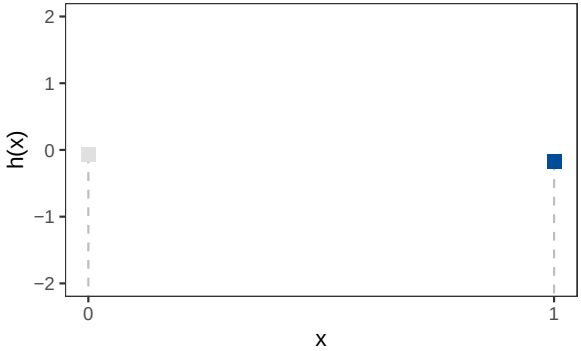
Since the domain of any $h(.) \in \mathcal{H}$ has only n elements, we can represent the function $h(.)$ compactly as a n-dimensional vector

$$\boldsymbol{h} = \left[h\left(\mathbf{x}^{(1)}\right), \ldots, h\left(\mathbf{x}^{(n)}\right)\right].$$

DISCRETE FUNCTIONS

Example 1: Let us consider $h : \mathcal{X} \to \mathcal{Y}$ where the input space consists of **two** points $\mathcal{X} = \{0, 1\}$.

Examples for functions that live in this space:



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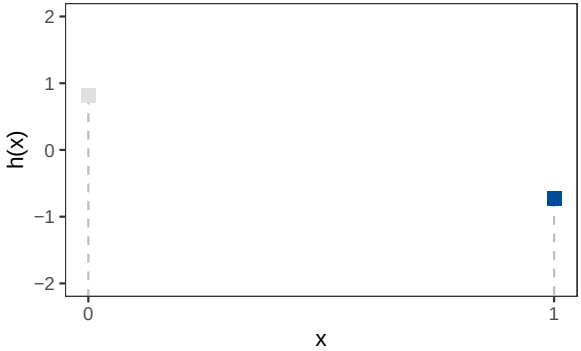

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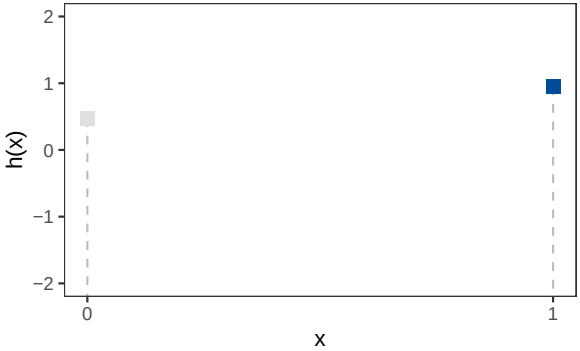

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Example 2: Let us consider $h : \mathcal{X} \to \mathcal{Y}$ where the input space consists of **five** points $\mathcal{X} = \{0, 0.25, 0.5, 0.75, 1\}$.

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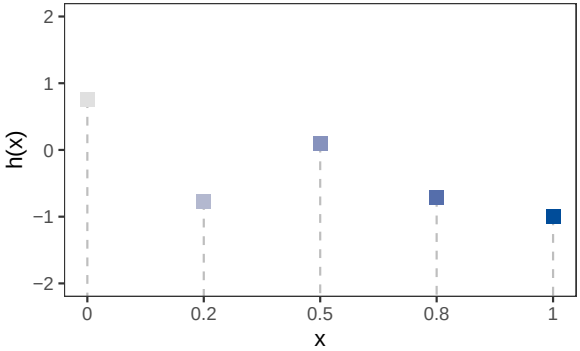

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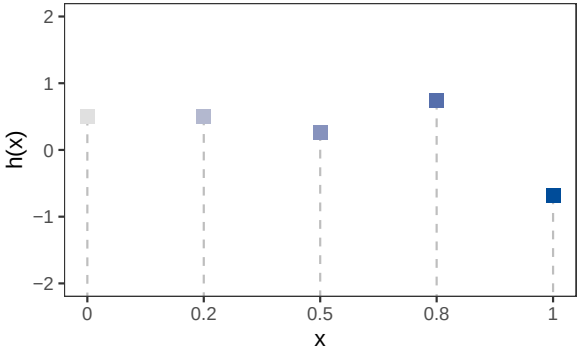

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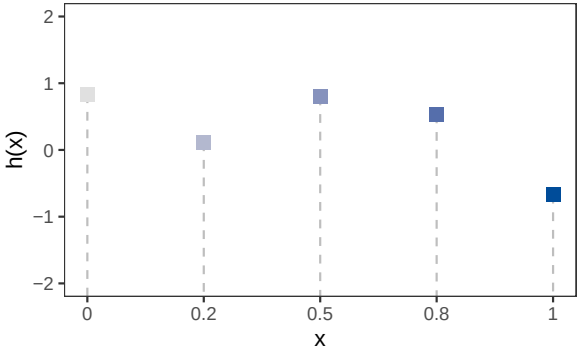

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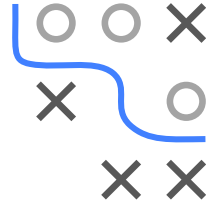

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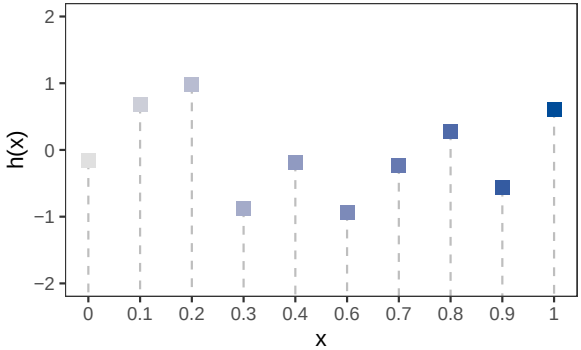

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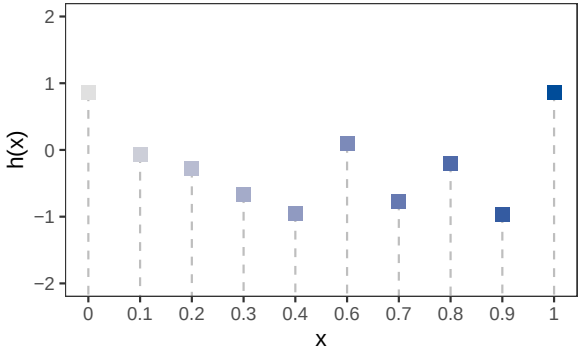

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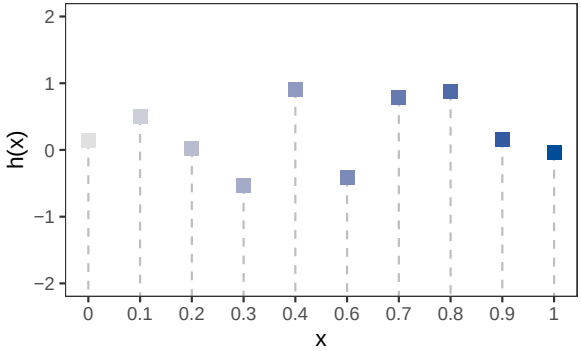
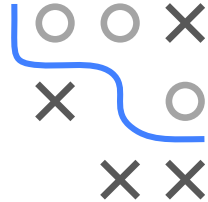


DISTRIBUTIONS ON DISCRETE FUNCTIONS



One natural way to specify a probability function on discrete function $h \in \mathcal{H}$ is to use the vector representation

$$\boldsymbol{h} = \left[h\left(\mathbf{x}^{(1)}\right), h\left(\mathbf{x}^{(2)}\right), \ldots, h\left(\mathbf{x}^{(n)}\right) \right]$$

of the function.

Let us see $\boldsymbol{h}$ as a n-dimensional random variable. We will further assume the following normal distribution:

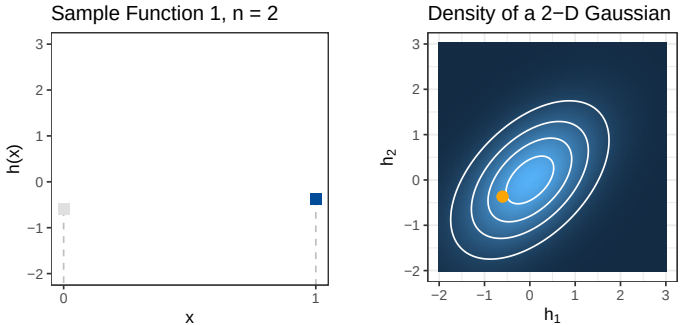
$$\boldsymbol{h} \sim \mathcal{N}\left(\boldsymbol{m}, \boldsymbol{K}\right).$$

Note: For now, we set $\boldsymbol{m} = \mathbf{0}$ and take the covariance matrix $\boldsymbol{K}$ as given. We will see later how they are chosen / estimated.

DISCRETE FUNCTIONS

Example 1 (continued): Let $h : \mathcal{X} \to \mathcal{Y}$ be a function that is defined on **two** points $\mathcal{X}$. We sample functions by sampling from a two-dimensional normal variable

$$\boldsymbol{h} = [h(1), h(2)] \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K})$$



In this example, $m = (0, 0)$ and $K = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$.

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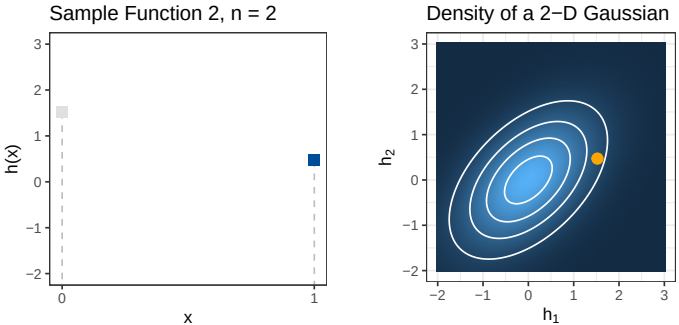

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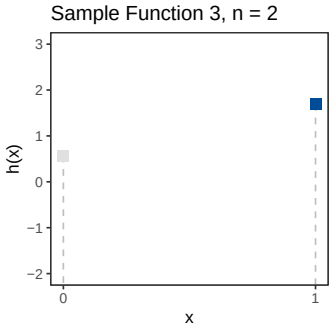

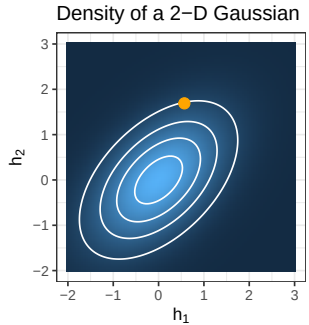

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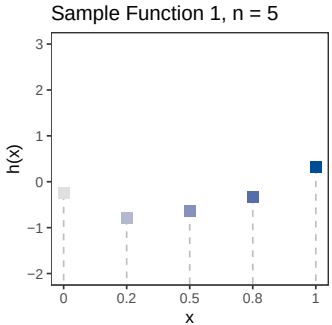

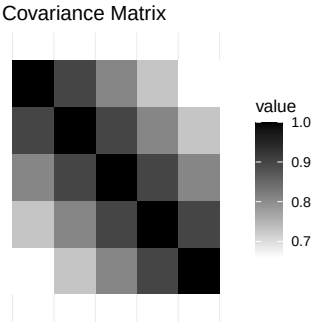

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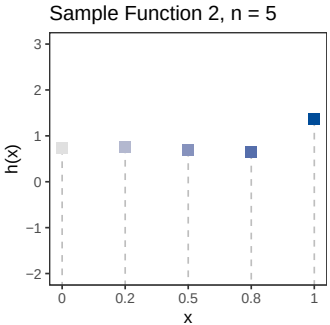

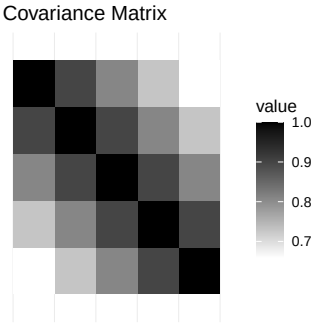

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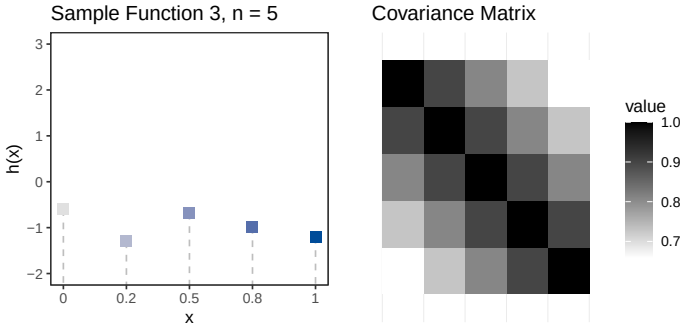

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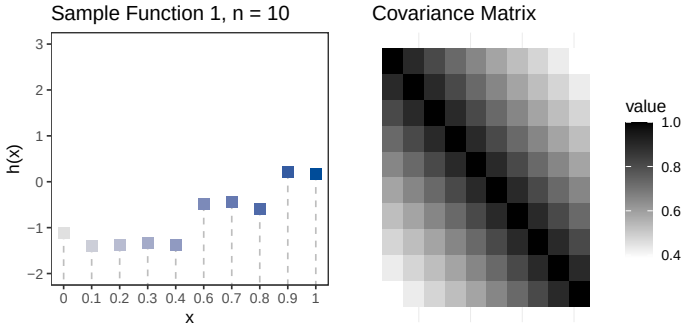

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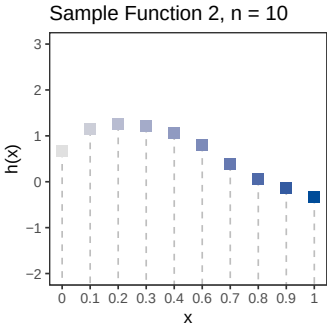

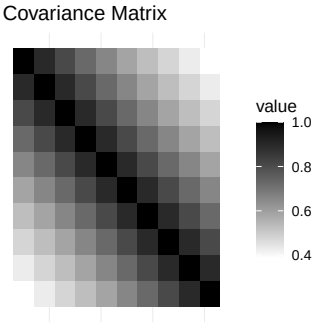

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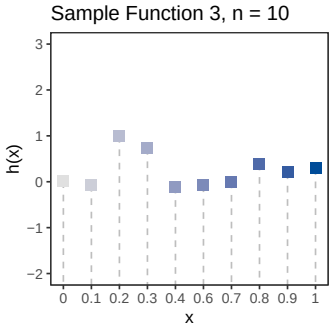
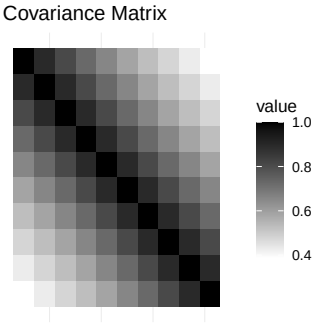


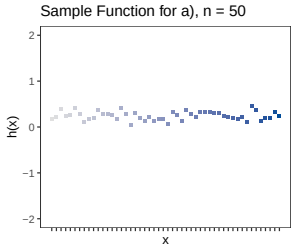


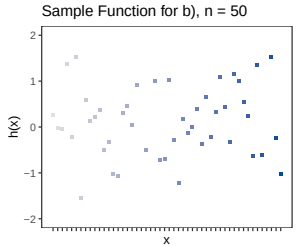
ROLE OF THE COVARIANCE FUNCTION

Note that the covariance controls the “shape” of the drawn function.
Consider two extreme cases where function values are

a) strongly correlated: $\boldsymbol{K} = \begin{pmatrix} 1 & 0.99 & \dots & 0.99 \\ 0.99 & 1 & \dots & 0.99 \\ 0.99 & 0.99 & \ddots & 0.99 \\ 0.99 & \dots & 0.99 & 1 \end{pmatrix}$

b) uncorrelated: $\boldsymbol{K} = \boldsymbol{I}$





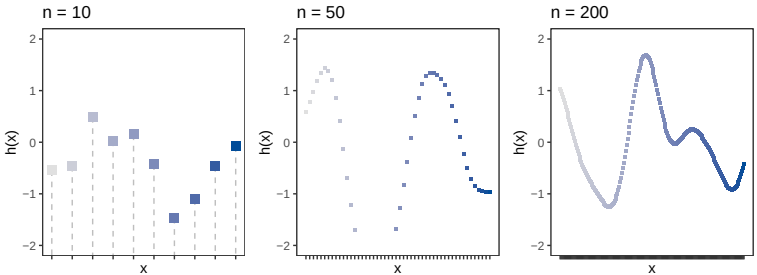
Gaussian Processes

FROM DISCRETE TO CONTINUOUS FUNCTIONS

- We defined distributions on functions with discrete domain by defining a Gaussian on the vector of the respective function values

$$\mathbf{h} = [h(\mathbf{x}^{(1)}), h(\mathbf{x}^{(2)}), \ldots, h(\mathbf{x}^{(n)})] \sim \mathcal{N}(\boldsymbol{m}, \boldsymbol{K})$$

- We can do this for $n \to \infty$ (as "granular" as we want)



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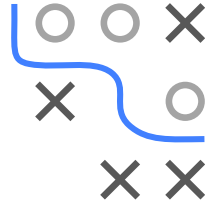


- No matter how large n is, we are still considering a function over a discrete domain.
- How can we extend our definition to functions with **continuous domain** $\mathcal{X} \subset \mathbb{R}$?

GAUSSIAN PROCESSES: INTUITION

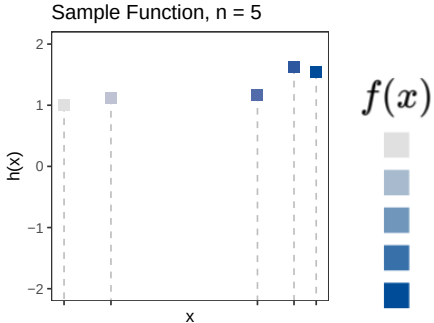
- Intuitively, a function f drawn from **Gaussian process** can be understood as an “infinite” long Gaussian random vector.
- It is unclear how to handle an “infinite” long Gaussian random vector!

GAUSSIAN PROCESSES: INTUITION

- Thus, it is required that for **any finite set** of inputs $\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\} \subset \mathcal{X}$, the vector $\mathbf{f}$ has a Gaussian distribution

$$\boldsymbol{f} = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}\left(\boldsymbol{m}, \boldsymbol{K}\right),$$

  with $\boldsymbol{m}$ and $\boldsymbol{K}$ being calculated by a mean function $m(.)$ / covariance function $k(.,.)$.
- This property is called **Marginalization Property**.



$$f(x) \sim \mathcal{N}\left(\mu, \boldsymbol{\Sigma}\right)$$

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$$\boldsymbol{f} = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}\left(\boldsymbol{m}, \boldsymbol{K}\right),$$

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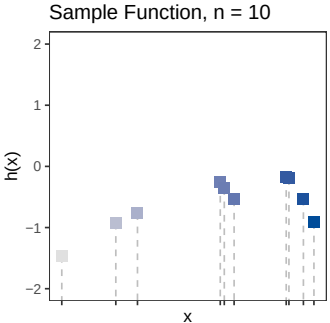
- This property is called **Marginalization Property**.



$$\sim \mathcal{N}\left(\mu, \boldsymbol{\Sigma}\right)$$

GAUSSIAN PROCESSES: INTUITION

- Thus, it is required that for **any finite set** of inputs $\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\} \subset \mathcal{X}$, the vector $\mathbf{f}$ has a Gaussian distribution

$$\boldsymbol{f} = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}\left(\boldsymbol{m}, \boldsymbol{K}\right),$$

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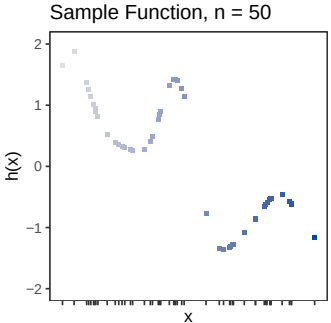
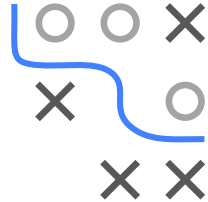


$$f(x)$$

$$\sim \mathcal{N}\left(\mu, \boldsymbol{\Sigma}\right)$$

GAUSSIAN PROCESSES



This intuitive explanation is formally defined as follows:

A function $f(\mathbf{x})$ is generated by a GP $\mathcal{GP}\left(m(\mathbf{x}), k\left(\mathbf{x}, \mathbf{x}'\right)\right)$ if for **any finite** set of inputs $\left\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right\}$, the associated vector of function values $\boldsymbol{f} = \left(f(\mathbf{x}^{(1)}), \ldots, f(\mathbf{x}^{(n)})\right)$ has a Gaussian distribution

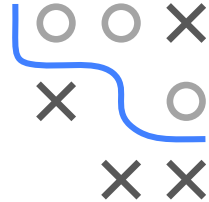
$$\boldsymbol{f} = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}\left(\boldsymbol{m}, \boldsymbol{K}\right),$$

with

$$\mathbf{m} \quad := \quad \left(m\left(\mathbf{x}^{(i)}\right)\right)_i, \quad \mathbf{K} := \left(k\left(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}\right)\right)_{i,j},$$

where $m(\mathbf{x})$ is called mean function and $k(\mathbf{x}, \mathbf{x}')$ is called covariance function.

SAMPLING FROM A GAUSSIAN PROCESS PRIOR



We can draw functions from a Gaussian process prior. Let us consider $f(\mathbf{x}) \sim \mathcal{GP}\left(0, k(\mathbf{x}, \mathbf{x}')\right)$ with the squared exponential covariance function [(*)]

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{1}{2\ell^2}\|\mathbf{x} - \mathbf{x}'\|^2\right), \quad \ell = 1.$$

This specifies the Gaussian process completely.

[(*)] We will talk later about different choices of covariance functions.

Gaussian Processes as Indexed Family

GAUSSIAN PROCESSES AS AN INDEXED FAMILY

A Gaussian process is a special case of a **stochastic process** which is defined as a collection of random variables indexed by some index set (also called an **indexed family**). What does it mean?

An **indexed family** is a mathematical function (or "rule") to map indices $t \in T$ to objects in $\mathcal{S}$.

Definition

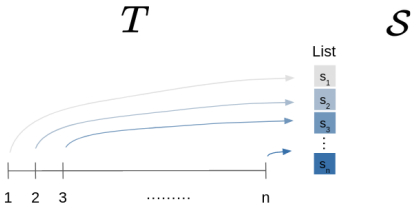
A **family of elements in $\mathcal{S}$ indexed by T** (indexed family) is a surjective function

$$\begin{aligned} s : T &\to \mathcal{S} \\ t &\mapsto s_t = s(t) \end{aligned}$$

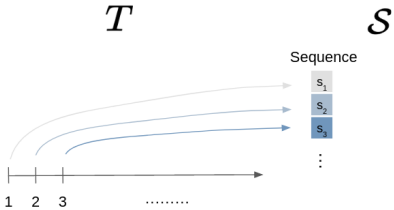
INDEXED FAMILY

Some simple examples for indexed families are:

- finite sequences (lists): $T = \{1, 2, \ldots, n\}$ and $(s_t)_{t \in T} \in \mathbb{R}$

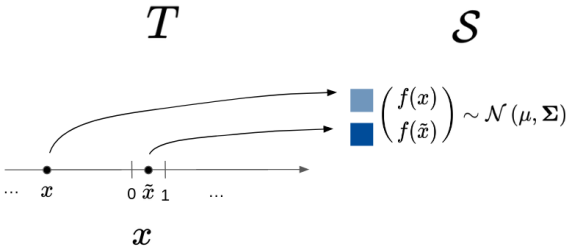


- infinite sequences: $T = \mathbb{N}$ and $(s_t)_{t \in T} \in \mathbb{R}$



INDEXED FAMILY

- A Gaussian process is also an indexed family, where the random variables $f(\mathbf{x})$ are indexed by the input values $\mathbf{x} \in \mathcal{X}$.
- Their special feature: Any indexed (finite) random vector has a multivariate Gaussian distribution (which comes with all the nice properties of Gaussianity!).



Visualization for a one-dimensional $\mathcal{X}$.

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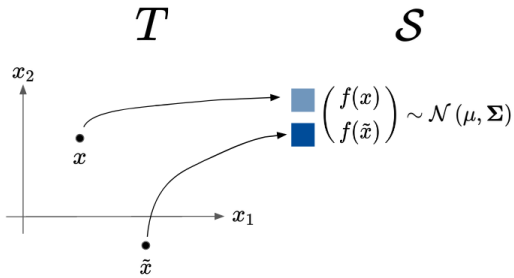

Visualization for a two-dimensional $\mathcal{X}$.